KOCHI METRO RAIL LIMITED

OPERATIONS & SAFETY DIVISION — MUTTOM OCC

REF: KMRL/OPS/SOP/2026/042
DATE: 24 October 2026
STATUS: OFFICIAL DRAFT

STANDARD OPERATING PROCEDURE: MONSOON READINESS & ELEVATED VIADUCT SAFETY DIRECTIVE (2026–2027)

Mandatory Compliance Guidelines for Train Operators, OCC Controllers & P-Way Engineers

Document ID: KMRL-SOP-2026-042	**Security Level:** Restricted / Departmental	**Revision:** 3.2 (Final)
Issuing Authority: Director of Rolling Stock	**Corridor:** Aluva – Petta – SN Junction	**Effective Date:** Immediate Effect

1. Executive Summary & Purpose

In accordance with Kerala State Disaster Management Authority (KSDMA) seasonal advisories and the Metro Railways (Operation and Maintenance) Act, this directive establishes fail-safe protocols governing metro train operations, viaduct water drainage, automatic signaling attenuation, and 750V DC traction isolation across all 25 operational stations.

2. Severe Weather Speed Restrictions & Automatic Train Protection (ATP)

Train operators and automatic driving modes must dynamically adhere to the following precipitation thresholds:

Weather Condition	Visibility Range	Operating Mode	Max Speed
Normal / Light Showers	> 500 meters	Auto Mode (ATO)	80 km/h
Moderate Monsoon Rain	200m – 500m	Coded Manual (CM)	60 km/h
Heavy Downpour / Squall	100m – 200m	Restricted Manual (RM)	45 km/h
Severe Storm (Wind > 70 km/h)	< 100 meters	OCC Caution Order	25 km/h

3. 750V DC Third Rail Traction & Drainage Integrity

- **Scupper Drain Inspections:** All viaduct downpipes and pier drainage basins between Kalamassery and Muttom depot must be inspected every 48 hours to prevent standing water.
- **Third Rail Clearance:** If standing water reaches within 50mm of the bottom shroud of the 750V DC bottom-contact third rail conductor, OCC must immediately execute controlled traction isolation.
- **ATC Balise Verification:** Post-flash flooding, signal engineers must execute transmission loopback tests on balise markers at turnaround sidings prior to passenger re-entry.

4. Emergency Contact Matrix

Department / Authority	Designation	Emergency Direct Line	Radio Channel

OCC Muttom Control Room	Chief Train Controller	+91 484 2846700	Ch 01 (Primary)
Permanent Way (P-Way)	Executive Engineer	+91 484 2846722	Ch 04 (Civil)
Traction Power Supply (TPS)	Power Controller	+91 484 2846744	Ch 02 (Electrical)
Disaster Management Unit	Incident Commander	1800 425 0355 (Toll Free)	Ch 08 (Emergency)

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